

DISTAL SENSOR INTERFACE LAYERS, SEALING, AND COUPLING EMBODIMENTS

In various embodiments, the distal sensing region of the intraluminal compliance and distance profiling device incorporates interface layers configured to protect sensing components while maintaining reliable ranging performance in moist, variable-reflectivity, and mechanically dynamic environments. Interface layers may improve repeatability of optical or acoustic measurements and may reduce measurement noise associated with fluid films, tissue contact variability, or contamination.

For optical time-of-flight or LiDAR embodiments, the distal sensing tip may include an optically transmissive window positioned over the emitter and receiver elements. The window may be formed from polymer, glass, sapphire, or other transparent materials and may be sealed to prevent fluid ingress. In certain embodiments, the window includes anti-fog coatings, hydrophobic or oleophobic coatings, anti-scratch coatings, or micro-textured surfaces configured to stabilize optical performance.

For ultrasonic embodiments, the distal sensing region may incorporate an acoustically transmissive membrane or coupling layer configured to transmit acoustic energy efficiently into the surrounding medium. Coupling structures may include soft membranes, gels, elastomeric caps, or integrated coupling chambers designed to improve acoustic impedance matching and reduce air-gap artifacts.

In certain embodiments, the distal tip includes a compliant protective cap that both protects sensing components and provides controlled contact behavior. Compliant tip structures may reduce localized pressure concentration while stabilizing sensor alignment relative to surrounding tissue.

Sealing embodiments may include bonded membranes, overmolded encapsulation layers, gaskets, O-rings, welded films, or hermetic housings preventing liquid intrusion into sensor cavities. Sealing may be applied to distal windows, internal cavities, cable passages, and interfaces between insertable sections and handle electronics.

In some implementations, the device includes venting or pressure-equalization pathways configured to prevent trapped pressure differentials from deforming membranes or windows. Such pathways may include one-way valves, hydrophobic vents, or controlled microchannels isolated from user contact surfaces.

Distal interface layers may be designed as replaceable components to support hygiene, maintenance, or performance tuning. Replaceable windows or caps may be exchanged to restore optical clarity, renew coatings, or adapt coupling behavior for different sensing modes.

The computational system may further account for interface-layer effects through calibration coefficients compensating for known window thickness, membrane compliance, acoustic propagation characteristics, or coating behavior. Such compensation improves consistency across devices and manufacturing lots.

Accordingly, distal sensor interface, sealing, and coupling embodiments enable reliable distance sensing under real-world intraluminal conditions and provide structural protection, maintenance flexibility, and performance stability across optical and ultrasonic sensing implementations.